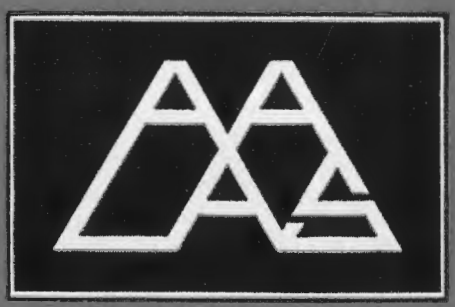
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THE YUCATAN MINIATURE PIG AS A LABORATORY ANIMAL[1,2]

L. M. PANEPINTO, R. W. PHILLIPS, L. R. WHEELER, AND D. H. WILL

SUMMARY • *Parent stock for a minature pig herd was obtained from the progeny of pigs imported from the Yucatan Peninsula of Mexico. The Yucatan miniature pig is slate-gray, essentially hairless, and has a docile temperament. The tractable nature of this pig minimized risks for animal handlers as well as researchers. With regular human contact, these pigs became amenable to handling and tolerated restraint well.*

The mean birth weight was 0.74 kg, and the mean weaning weight at 60 days was 6.47 kg. The mean weight of mature nonobese boars was 83 kg while mature sows averaged 70 kg. The height and length for mature pigs of both sexes averaged 57 cm and 76 cm, respectively. The mean range of plasma glucose was 86–117 mg/dl during the first 8 weeks of life.

The Yucatan miniature pig is native to tropical regions. The Colorado cold winter weather caused life-threatening stress to animals housed outside; therefore, intensive management procedures were required during the winter months.

KEY WORDS • *Breeding—Growth—Yucatan miniature pig*

GENERAL CHARACTERISTICS

The Yucatan miniature pig, which is native to southern Mexico and Central America, has been developed for use in biomedical research. The animal is called the Mexican hairless pig (1). Parent stock for the herd was obtained from the progeny of pigs imported from the Yucatan Peninsula of Mexico near Frontera. The Yucatan miniature pig is slate-gray and essentially hairless; it is very tractable. A number of animals have wattles on the lower neck (Figure 1).

The absence of a hair coat facilitates a number of laboratory procedures and reduces the heavy odor found in most breeds of swine. For example, extensive shaving for surgery or blood vessel location is unnecessary. Also, the general cleanliness of these animals and their facilities is maintained more easily, especially under confinement or laboratory conditions. The Yucatan miniature pig is less inclined than other breeds to practice rooting or digging when maintained outside on dirt.

A dramatic differentiation in skin occurs between males and females, with boars developing an armor-like plate over the shoulder region. The thickening of the skin occurs primarily in the dermis. The keratinized epithelium is of normal histological appearance. At maturity, the skin may reach a thickness of over 3 cm. The sows develop and maintain a much softer, more pliable skin. This differentiation was not mentioned in an early paper dealing with the histology of the skin and follicular development of this breed (2).

For short-term restraint (up to 5 hours), a reinforced canvas sling apparatus has been developed. The sling is attached to a portable steel frame and allows easy access to the precava, cephalic vein, or ear vessels while holding the animal in a comfortable position (Figure 2).

[1] From the Department of Physiology and Biophysics, Colorado State University, Fort Collins, CO 80523.
[2] Supported by a grant from the Kroc Foundation.



Fig 1. *Head profile of Yucatan miniature swine. This pig is hairless and has wattles on lower neck.*

For longer term studies requiring blood sampling via an implanted catheter, best results have been obtained by confining but not restraining catheterized animals in a small pen or farrowing crate. This is also of benefit for studies in which physical restraint might modify testing results. The Yucatan miniature pig rapidly becomes accustomed to close contact and handling, and blood samples can be taken with minimal stress to the pigs.

Breeding Characteristics

Average litter size (live births) for both gilts and sows in the Colorado State University herd is 6.12 ± 1.78 (standard deviation). Sows in this herd average 6.22 ± 2.13 while gilts average 5.96 ± 1.03 pigs per litter.

Normal estrous cycle is 20 days with a gestation period of 110–115 days. Sexual maturity is comparable to that of domestic



Fig 2. *Canvas sling on portable steel frame for restraint of Yucatan miniature swine.*

breeds (3), occurring at 4–7 months of age, but mating of gilts at that age is not recommended primarily due to their small size (approximately 20 kg at 6 months of age).

Breeding of Yucatan miniature swine poses no special problems, and sows readily accept a boar when in estrus. When sows are housed outside, physical and behavioral signs of estrus are not observed as frequently during the winter months. Further, when gilts are raised to puberty in an indoor facility, onset of first estrus may be delayed beyond 7 months of age. Handling of boars does not require special facilities or equipment, and the sometimes dangerous nature of intact males previously reported in other miniature swine breeds (4) has not been observed in the Yucatan board. The absence of hostility toward handlers is predictable with few exceptions whether boars are housed individually, in groups, or with sows during breeding season. The use of hog snares or nose rings has never been practiced in this colony.

When adult boars are penned in large groups, reintroduction of individuals temporarily removed from the pen can result in injury or death to the reintroduced boar. However, in groups of five or less this has not been a problem, nor does this difficulty occur when the boar being moved is considerably smaller in size than those of the main group. Occasionally, an older adult boar that has been housed alone for 1–2 years after puberty will be intolerant of other males. In order to avoid this type of conflict, housing boars in groups of at least two per pen is recommended and should be initiated within the first year of life. Further, careful supervision upon introduction of a new boar to a group should be maintained.

Farrowing should be attended for maximum piglet survival, and the use of a farrowing crate at parturition is recommended. In most cases, they can be farrowed while free in a small pen. Crushing of piglets is not a serious problem unless sows are unattended at farrowing. Litters can be maintained without a paneled off area once the sow has completed delivery. This procedure has become routine in this laboratory when sufficient numbers of farrowing crates are unavailable. Most sows will allow handling of their young at birth and thereafter without the necessity of protective paneling or fences.

TABLE 1

Neonatal growth characteristics of 33 Yucatan miniature pigs

Age (days)	Shoulder height (cm)	Girth (cm)	Length (cm)	Weight (kg)
0	14.3 ± 0.9[a]	20.0 ± 1.6	30.1 ± 2.1	0.67 ± 0.09
1–3	15.9 ± 0.8	20.9 ± 1.6	30.7 ± 1.9	0.74 ± 0.11
7–10	18.2 ± 0.9	27.9 ± 1.4	38.1 ± 2.0	1.61 ± 0.20
11–15	18.6 ± 1.0	29.0 ± 1.3	38.4 ± 3.6	1.83 ± 0.22
16–21	18.8 ± 1.1	29.5 ± 1.4	42.0 ± 3.1	2.14 ± 0.33
22–28	21.4 ± 1.5	33.5 ± 2.1	47.2 ± 1.9	2.72 ± 0.85
29–34	22.4 ± 1.3	36.3 ± 2.8	51.3 ± 3.0	3.94 ± 0.47
35–40	23.6 ± 1.6	37.4 ± 3.3	52.9 ± 4.3	4.41 ± 1.00
41–48	23.9 ± 1.8	37.7 ± 2.9	54.3 ± 5.8	4.65 ± 1.24
49–56	26.4 ± 1.6	41.5 ± 2.3	60.9 ± 3.6	6.69 ± 0.83
57–66[b]	26.4 ± 4.7	44.4 ± 6.8	61.4 ± 7.9	7.24 ± 2.34

[a] Mean ± standard deviation
[b] Winter weather period

At birth, all pigs are weighed and ear notched. The needle teeth are clipped, and 100 mg of iron dextran is administered intramuscularly. Since they lack the characteristic hair coat of other swine breeds, these neonates are predisposed to cold stress, and maintenance of a warm environment is critical to their survival. Litters should be maintained inside for a minimum of 1 week after farrowing. If they are housed outside prior to weaning, they should be provided with an infrared heat lamp until they are weaned. Birth weights for both sexes average 0.74 ± 0.17 kg (n = 580), with an average weaning weight at 60 days of 6.47 ± 2.10 kg. This is slightly less than the average birth weight reported for the Hormel miniature swine, 0.87 ± 0.24 kg (5), and slightly greater than that observed in the Hanford miniature swine, 0.71 kg (1).

Growth Characteristics

The growth rate for a selected test group of 33 piglets representing six litters is shown in Table 1. Breeding stock consisted of six sows and five boars; all litters were farrowed inside under supervision. Each litter was sow-raised and kept inside a minimum of 7 days before being moved to an outside pen. Each outside pen measured 7.3 x 14.6 m and included a pig hut measuring 2.4 x 2.4 m with a heat lamp inside. Throughout gestation and lactation, sows were limit-fed the swine diet outlined in Table 2, and the piglets had available a standard pig starter[3] (17% protein) *ad libitum* until weaning.

The pigs were weighed and measured at birth and every other week thereafter until weaning with all measurements supervised by the same technician. Length was measured from the nose to the base of the tail, girth around the body at the level of the heart, and height from the floor to the top of the shoulder. Length, girth, and height measurements are summarized in Table 1. Average weight for males and females at 1 year of age is 49 ± 9 kg; full growth is attained at 20–24 months. Nonobese mature boars have a mean weight of 83 ± 12 kg, ranging from 59 to 105 kg, while sows at maturity average 70 ± 12 kg, ranging from 40 to 90 kg. This is somewhat less than the 2 to 3-year-old mature weight of Hormel miniature swine (90–136 kg) (6) and comparable to that of Pitman-Moore miniature swine (males 79 kg, females 70 kg) (7).

Females show a greater tendency for obesity than males when fed *ad libitum*, and obese sows have reached weights in excess

[3] Ranch-way Pig Starter. Ranch-way Mills, Ft Collins, CO.

TABLE 2

All-purpose diet composition for Yucatan miniature pigs

Amount (kg)	Ingredients
591	Corn (fine chop kernels)
136	Dehydrated alfalfa meal
114	Soybean oil meal
45	Wheat bran
9	Calcium carbonate
9	Dicalcium phosphate
4.5	Sodium chloride
2.3	Swine premix[a]
0.9	Mineral supplement[b]

[a] Pig MVP, Cargill Nutrena, Feed Division, Minneapolis, MN
[b] 12:12 mineral, Cargill Nutrena, Feed Division, Minneapolis, MN

of 140 kg. This weight gain in mature adults represents an excess of 50 kg over normal body weight. Mean height and length for mature Yucatan pigs of both sexes are 57 ± 4 cm and 76 ± 5 cm, respectively.

Blood samples were taken at birth and every other week, alternating between the two halves of the same group of neonates whose growth was characterized above. Blood was removed from the anterior vena cava using a 21-gauge needle; care was taken to minimize stress to the animals during sampling. Plasma glucose values reported in this study are comparable to those observed in the neonatal Hormel miniature swine (8) with a mean value ranging from 86 ± 20 to 117 ± 9 mg/dl during the first 8 weeks of life (Table 3). Blood glucose in neonatal Pitman-Moore miniature swine (9) was higher during the first 21 days (146 ± 22 to 124 ± 30 mg/dl), but had dropped to 71 ± 5 mg/dl by 56 days of age.

Plasma glucose level at birth, prior to nursing, was not reported in previous studies (8, 9). Zero-time blood samples in this report represent samples taken within 10 minutes of birth and before nursing. Mean plasma glucose for this time period in the Yucatan miniature pig was 51 ± 13 mg/dl. With the exception of the zero-time samples, all blood and serum determinations were made on nonfasting animals with time post-nursing unknown.

Packed cell volume in the Yucatan min-

TABLE 3

Neonatal hematology values of 33 Yucatan miniature pigs

Age (days)	Plasma glucose (mg/dl)	Plasma protein (g/dl)	Packed cell volume (%)	Serum cholesterol (mg/dl)	Serum cholesterol ester (mg/dl)	Serum phospholipids (mg/dl)	Serum triglycerides (mg/dl)
0 (birth)	51 ± 13[a]	3.6 ± 1.1	42 ± 4				
1–3	86 ± 20	4.3 ± 1.5	37 ± 6				
7–10	112 ± 16	7.7 ± 0.2	39 ± 2				
11–15	115 ± 11	6.3 ± 1.0	38 ± 5	136 ± 38	104 ± 27	12.8 ± 16.6	84 ± 2
16–21	117 ± 9	6.0 ± 0.6	40 ± 2	144 ± 18	116 ± 12	9.2 ± 1.0	93 ± 3
22–28	115 ± 9	5.9 ± 0.4	39 ± 5	132 ± 30	117 ± 21	8.6 ± 1.5	86 ± 1
29–34	112 ± 9	5.9 ± 0.3	40 ± 16	112 ± 26	98 ± 20	9.2 ± 1.7	69 ± 14
35–40	104 ± 12	6.2 ± 0.3	41 ± 5	142 ± 25	124 ± 20	10.0 ± 3.5	90 ± 22
41–48	102 ± 11	6.8 ± 0.8	42 ± 4	177 ± 60	153 ± 55	10.6 ± 3.0	83 ± 24
49–56	102 ± 16	6.4 ± 0.5	42 ± 5	224 ± 137	108 ± 15	6.7 ± 1.2	87 ± 9
57–66	99 ± 7	6.7 ± 0.7	42 ± 2	113 ± 6	109 ± 9	7.0 ± 0.6	103 ± 17

[a] Mean ± standard deviation

iature pig ranged from 41.8 ± 3.7% at birth to 47.3 ± 2.1% at 66 days with some fluctuations during the 2-month period. This is higher than that seen in Hormel miniature swine which had a 7-day measurement of 31.4 ± 3.0% and a 56-day packed cell volume of 37.1 ± 7.8% (8). However, the higher value observed in our swine may be a function of the altitude at which the colony is maintained (1,450 m). Serum cholesterol in Pitman-Moore miniature swine (9) was elevated as compared to our pigs during the first 2 months of life. Values of all hematological parameters measured are presented in Table 3.

Discussion

The docile, friendly temperament of the Yucatan miniature pig minimizes risk for animal handlers as well as researchers. With minimal but regular human contact, these pigs become amenable to handling and tolerate restraint quite well. The Yucatan pig also possesses tolerance toward others of its breed. Adult boars can be housed together or, if incompatible in the same pen, in adjacent pens separated by wire fencing with a minimum of hostility toward each other. Pigs of extreme size and age difference also are compatible. Sows with litters can be housed together with nonpregnant females, and adult boars show no hostility toward piglets even when confined together in a small pen.

The Yucatan pig is native to tropical regions. This, in conjunction with their characteristic hairlessness, can pose problems for researchers housing the pigs outside in areas where extreme seasonal climatic changes occur. Our winter weather is a life-threatening stress to pigs housed outside, and without intensive management, acute hypothermia or pneumonia may occur. A relatively warm (18–23°C), stable environment is essential for conducting experiments dealing with metabolic function.

The Yucatan miniature pig has shown some tendency toward rectal prolapse. However, the addition of 5% wheat bran to the diet has eliminated the prolapses in this herd. The diet, developed as an all-purpose growing, maintenance, and breeding diet, is presented in Table 2. The unique physical and behavioral characteristics of the Yucatan miniature pig make it a potentially valuable laboratory animal.

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